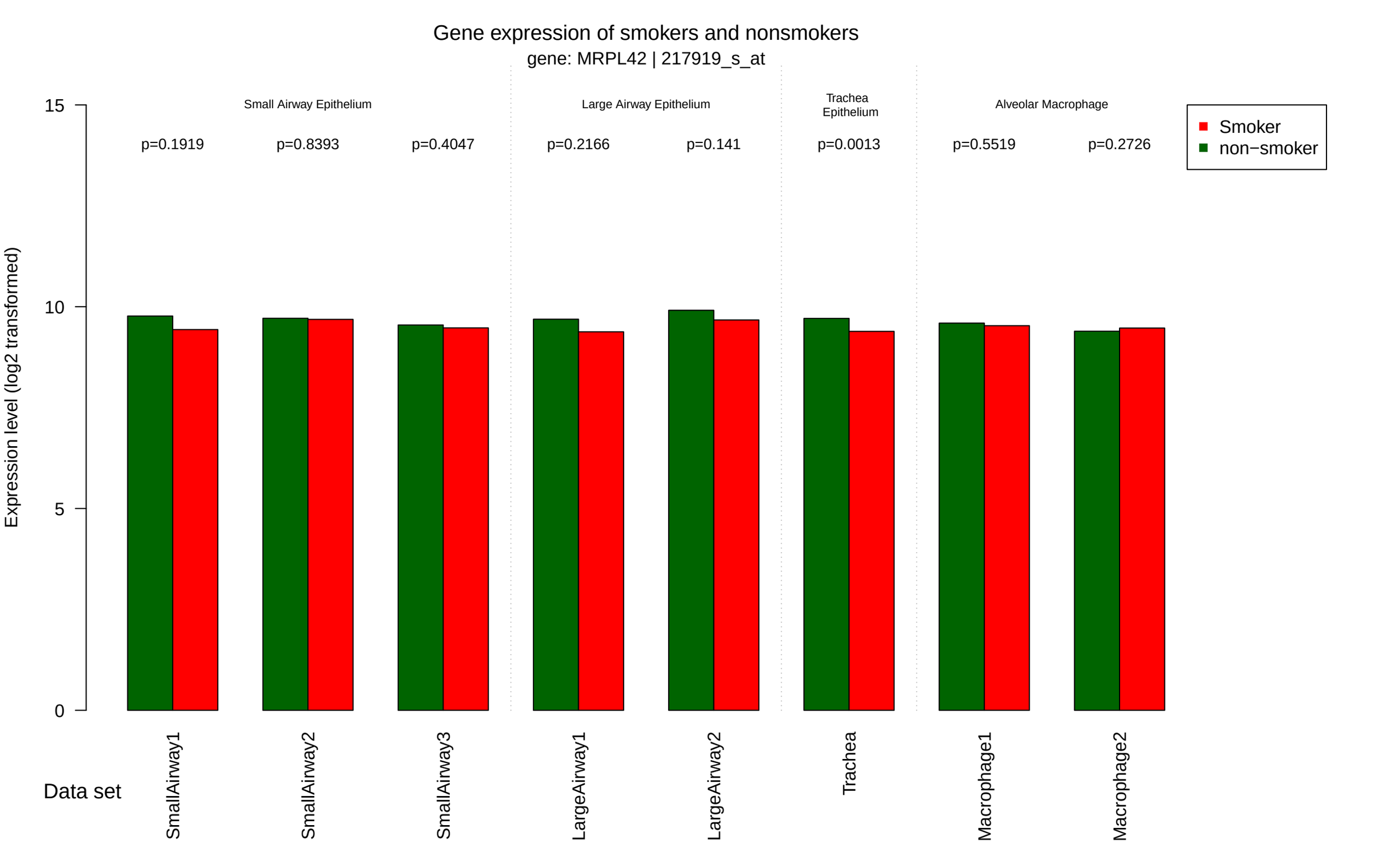
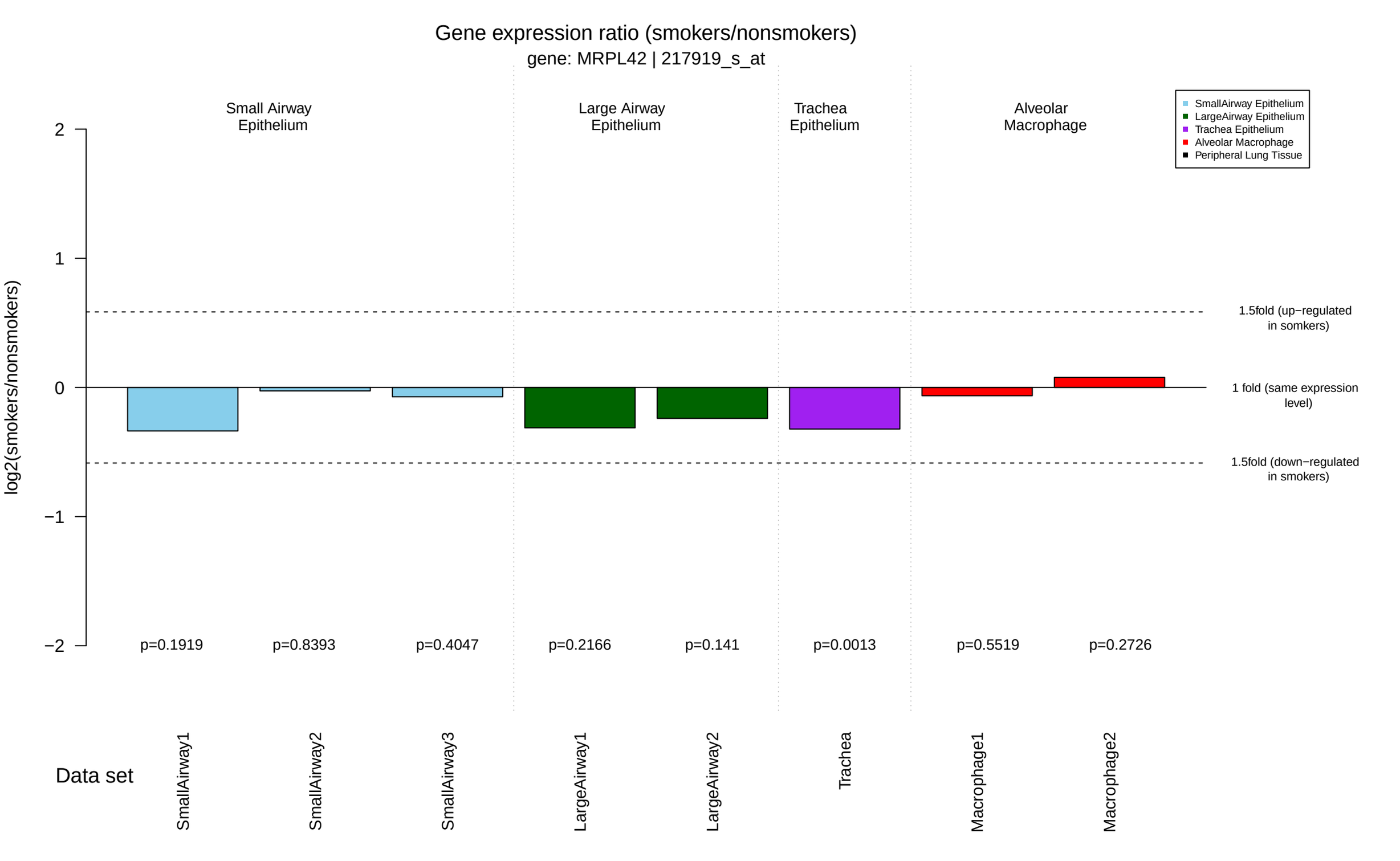
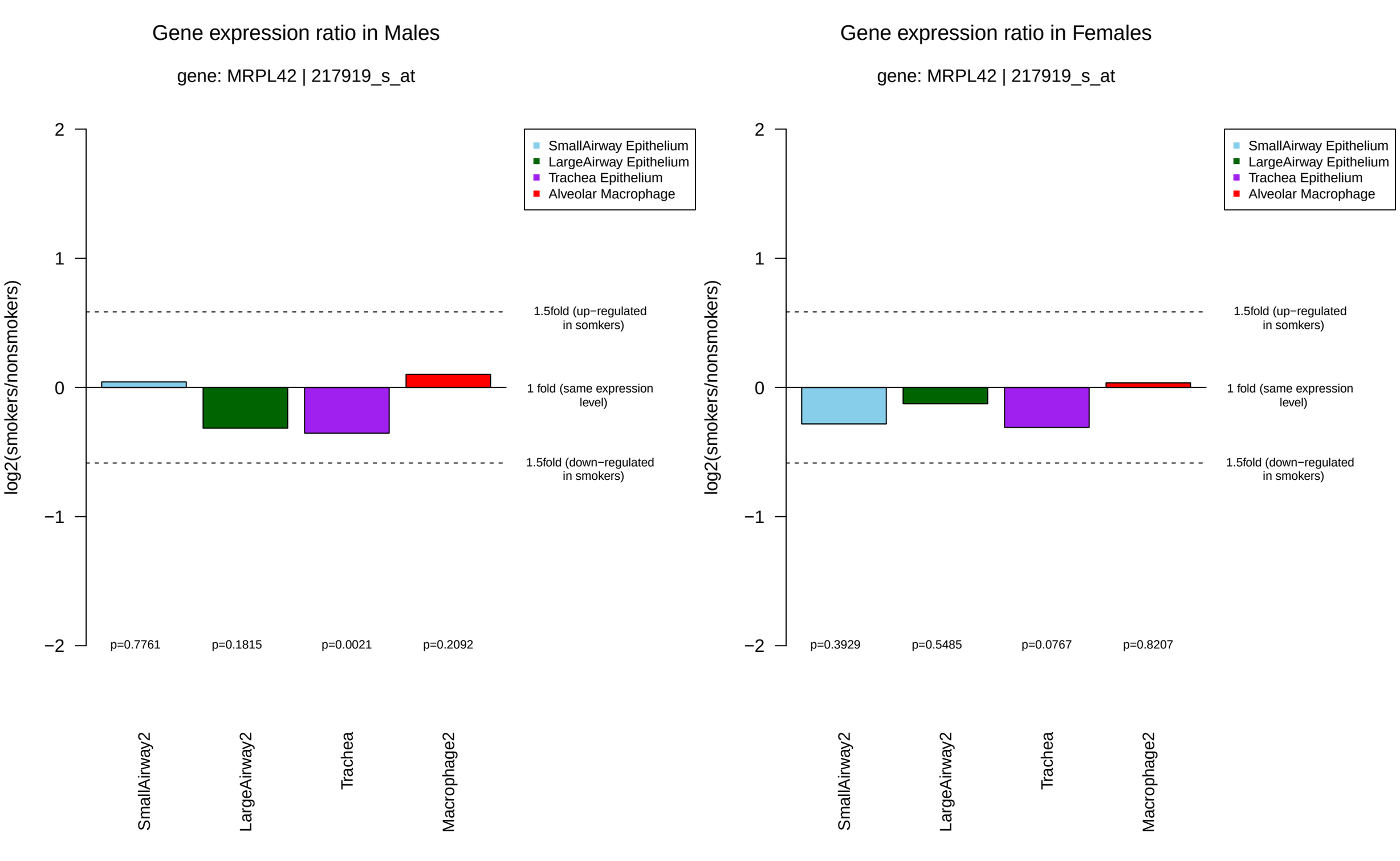
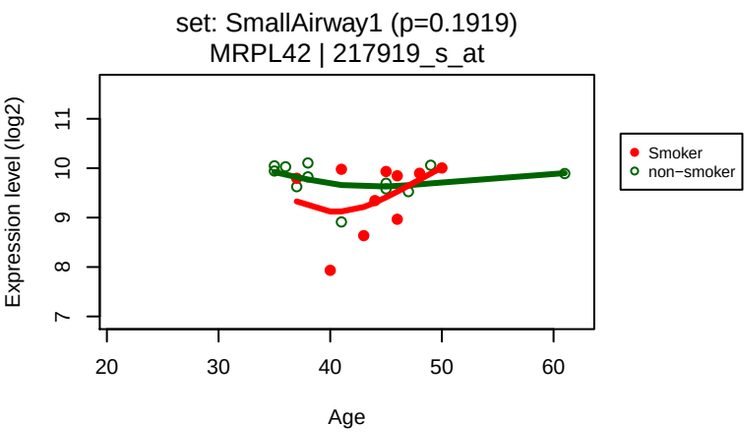
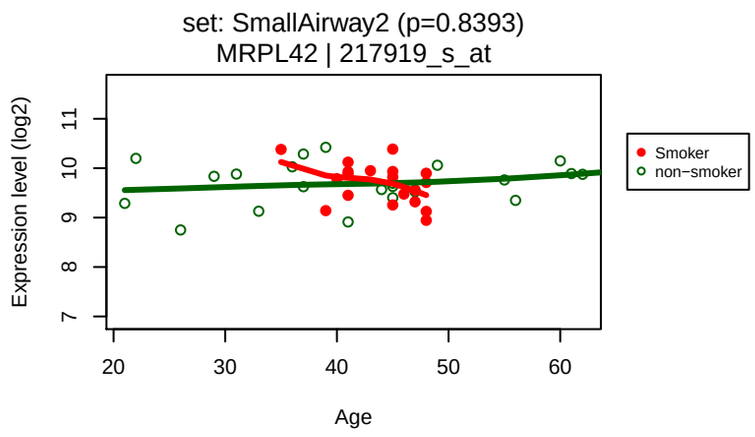
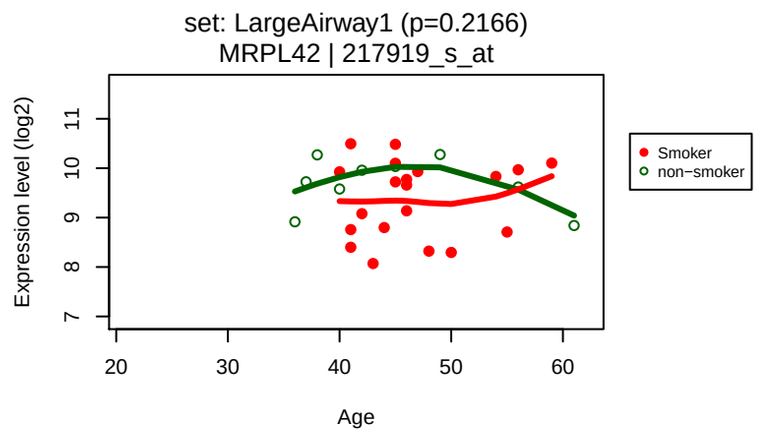

Gene expression of smokers and nonsmokers
gene: MRPL42 | 217919_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1919
p=0.8393
p=0.4047
p=0.2166
p=0.141
p=0.0013
p=0.5519
p=0.2726
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

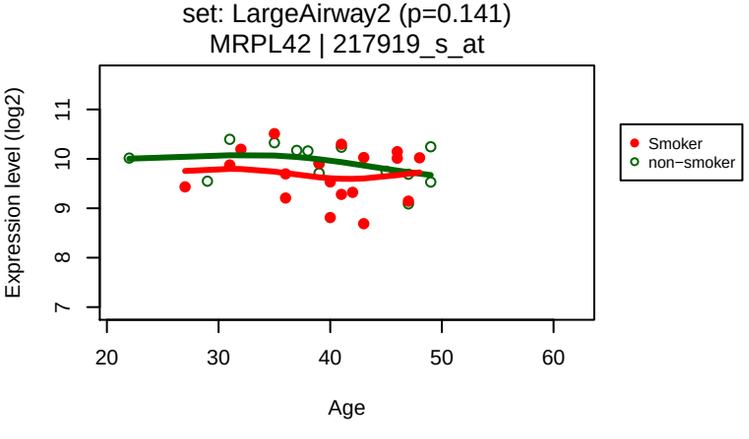

Gene expression ratio (smokers/nonsmokers)
gene: MRPL42 | 217919_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1919
p=0.8393
p=0.4047
p=0.2166
p=0.141
p=0.0013
p=0.5519
p=0.2726
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

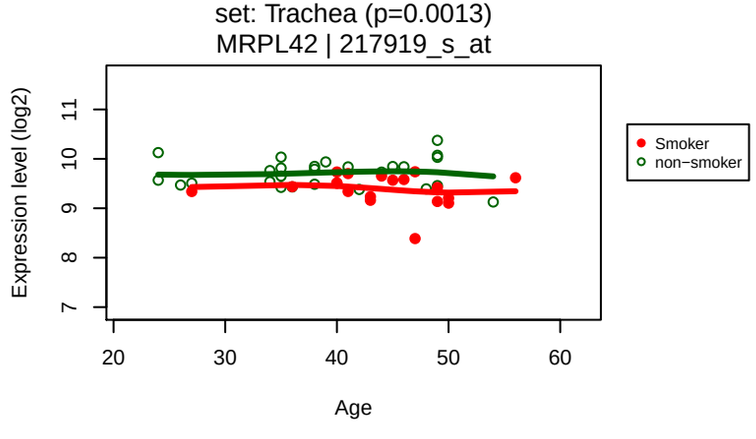

Gene expression ratio in Males
gene: MRPL42 | 217919_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7761
p=0.1815
p=0.0021
p=0.2092
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL42 | 217919_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3929
p=0.5485
p=0.0767
p=0.8207
SmallAirway2
LargeAirway2
Trachea
Macrophage2

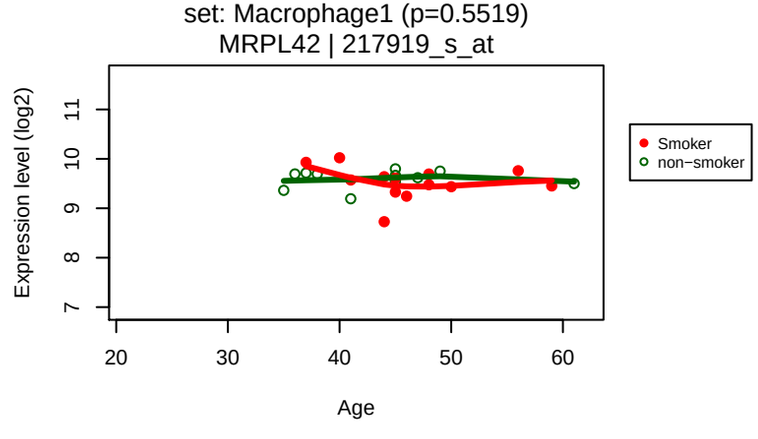

set: SmallAirway1 (p=0.1919)
MRPL42 | 217919_s_at
Expression level (log2)
Age
Smoker
non−smoker

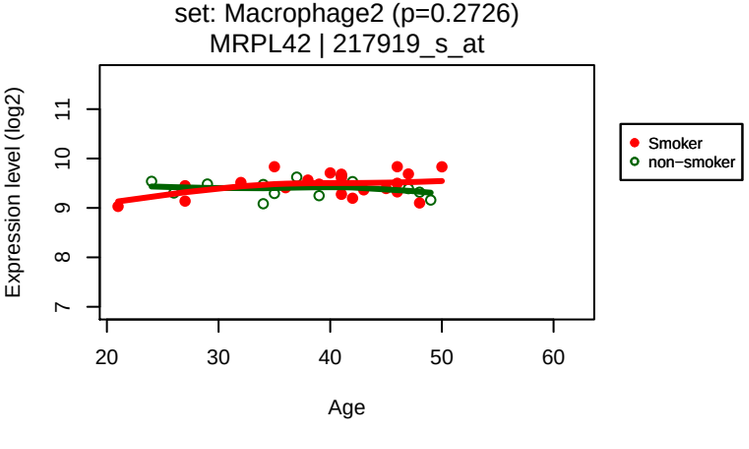

set: SmallAirway2 (p=0.8393)
MRPL42 | 217919_s_at
Expression level (log2)
Age
Smoker
non−smoker

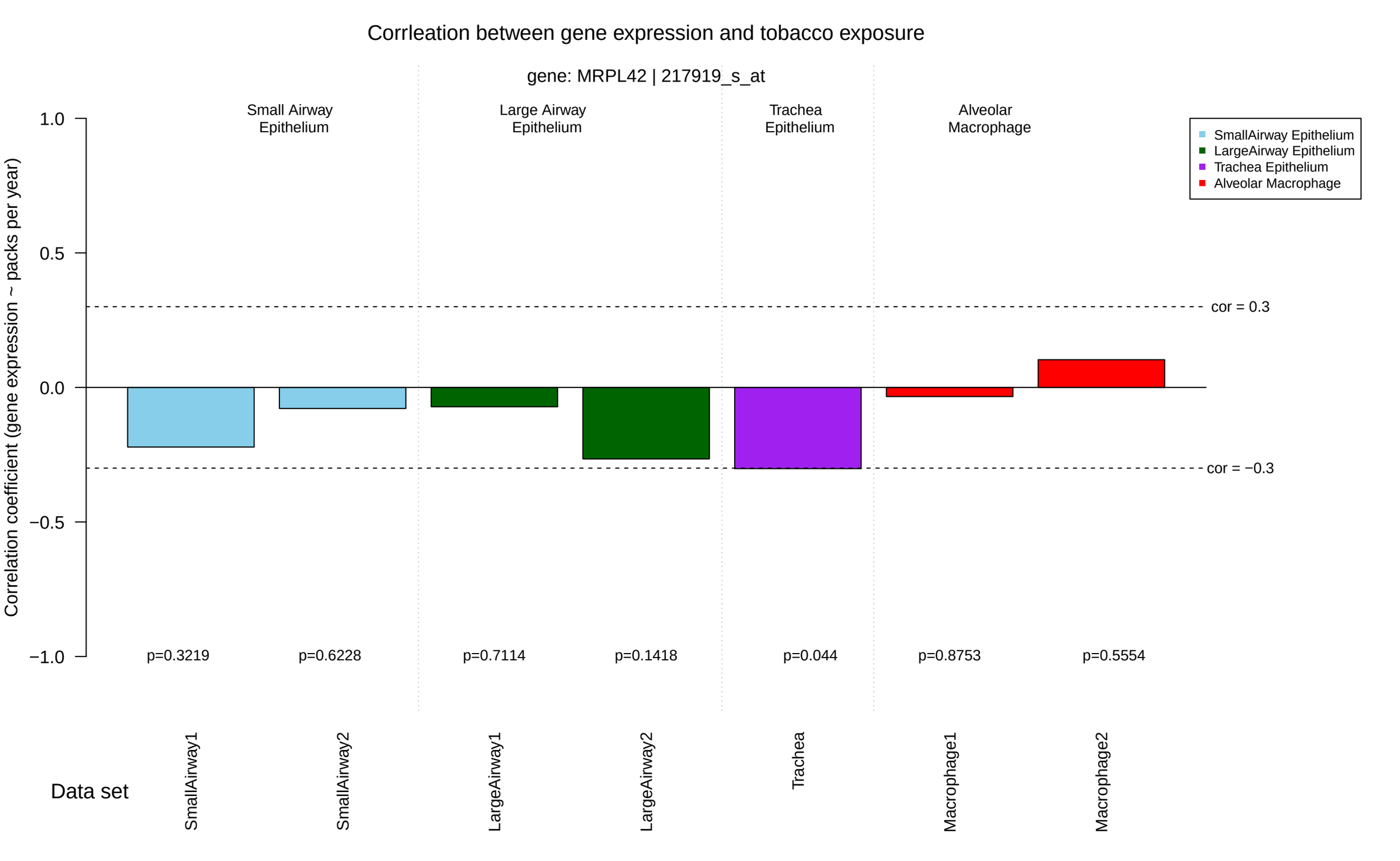

set: LargeAirway1 (p=0.2166)
MRPL42 | 217919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.141)
MRPL42 | 217919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
MRPL42 | 217919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5519)
MRPL42 | 217919_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2726)
MRPL42 | 217919_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL42 | 217919_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3219
p=0.6228
p=0.7114
p=0.1418
p=0.044
p=0.8753
p=0.5554
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2